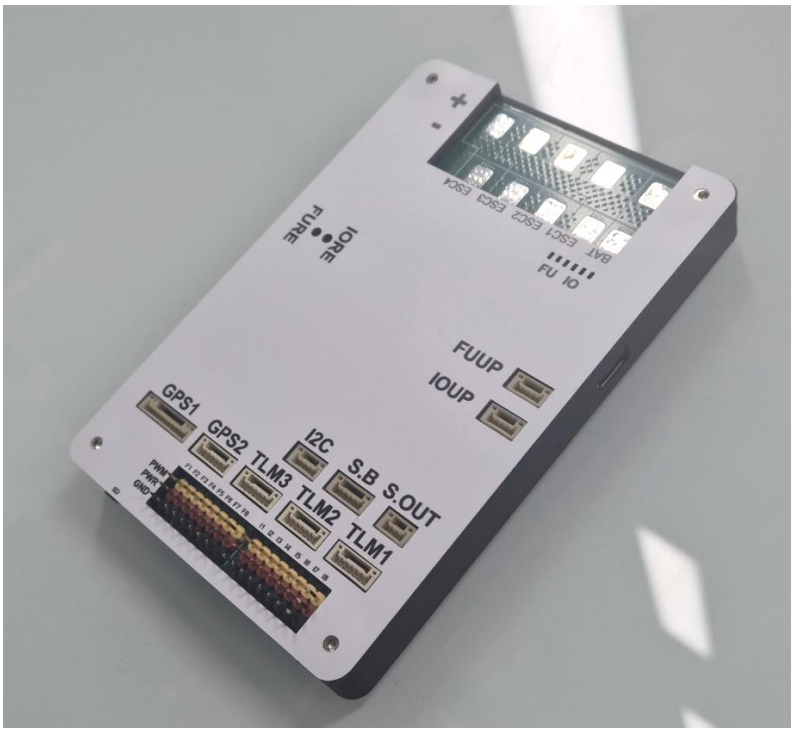
Corp.DCLASS

DCD FC

Integrated power distribution flight controller | Product and connection guide

DCD FC is a flight controller with integrated power distribution developed by Corp.DCLASS. It combines battery input and power connections for four external ESCs on one board. GPS, telemetry, receiver interfaces and 16 PWM outputs support aircraft integration.



DCD FC with enclosure

Specifications

Processors	STM32H743IIT6 main MCU / STM32F103C8T6 IO MCU
IMUs	BMI088 + ICM-42688-P accelerometers and gyroscopes
Barometer / compass	MS5611 / IST8310
Storage	FRAM parameters / microSD flight logs
Outputs / interfaces	16 PWM outputs, 2 GPS, 3 telemetry, I2C, SBUS input, USB
Input voltage	Up to 25.2 V (6S LiPo) / supports 3–6S
Power distribution	Power pads for 4 external ESCs / ESCs not included
Size / weight	Approx. 126 × 84 × 13 mm / 103 g with enclosure

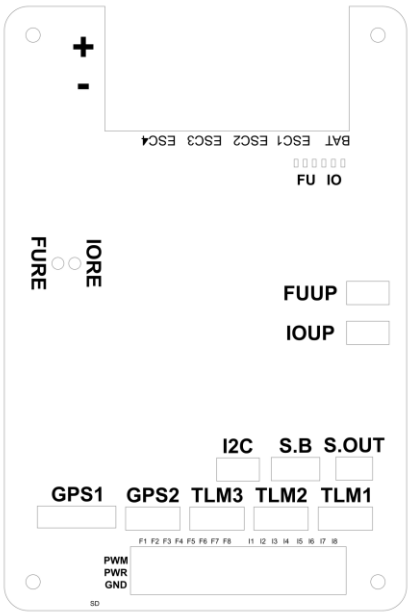
Dimensions refer to the enclosure, excluding protruding connectors and pins.

Purchase and technical support | Corp.DCLASS | https://www.d-class.co.kr

Connect to a PC using a standard USB data cable. Flight logs are stored on microSD.

Ports and connections

Use the top-panel labels and the table below to identify ports and their functions.



BAT / ESC1~4	Battery input / external ESC power
GPS1 / GPS2	GPS / SERIAL3 / SERIAL4
TLM1 / TLM2	Telemetry / SERIAL1 / SERIAL2
TLM3	Additional UART / SERIAL5
I2C	External compass and I2C peripherals
S.B	SBUS receiver input
F1~F8	FMU PWM / SERVO9–16
I1~I8	IO PWM / SERVO1–8
USB / SD	PC connection / microSD logs
FUUP / IOUP	FMU / IO programming via SWD

S.OUT carries the IO USART3 TX signal. FURE and IORE are reset controls; FU and IO are status indicators. PPM input and external CAN ports are not supported.

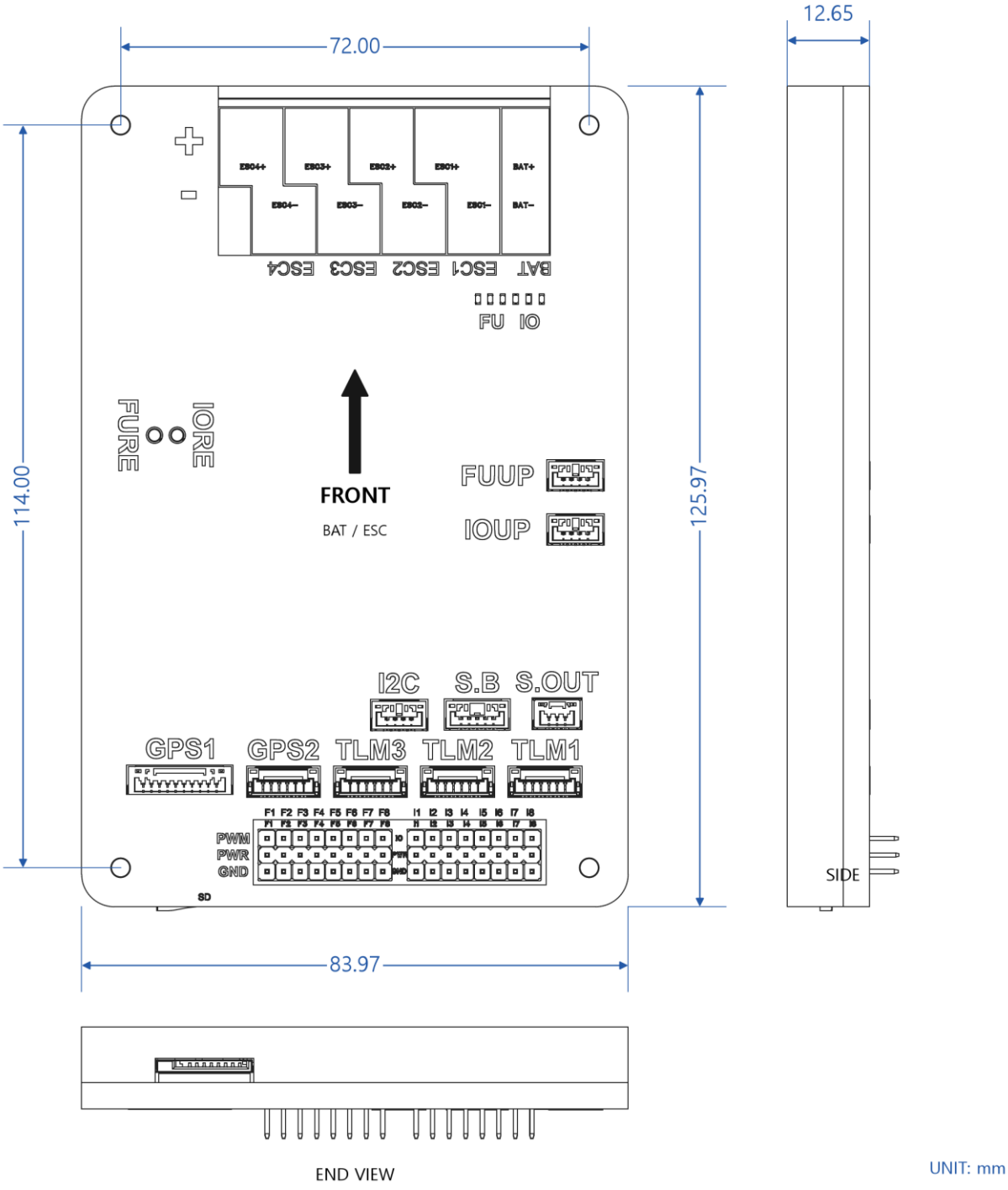
Port power supply

Port / power pin	Voltage	Supply current and sharing
GPS1·GPS2 / TLM1~3 / I2C / S.B	5 V	Up to 1.5 A total, including internal FC consumption.
F1–F8 + I1–I8 / PWR row	5 V	Up to 1.5 A total, shared by F1–F8 and I1–I8. Separate from the FC/peripheral supply.
GPS1 / pin 3	3.3 V	1 A supply rating, shared with microSD.
S.OUT	—	No power output pin.
FUUP / IOUP / pin 1	3.3 V REF	Debugger voltage reference; do not use to power external devices.
BAT / ESC1~4	Battery voltage	Battery distribution to external ESCs. Up to 25.2 V (6S LiPo).

Current figures apply to each shared supply as a whole. Allow for internal consumption and thermal limits. USB power is also subject to host and cable limits.

Dimensions and forward direction

Front points toward the BAT and ESC power pads.



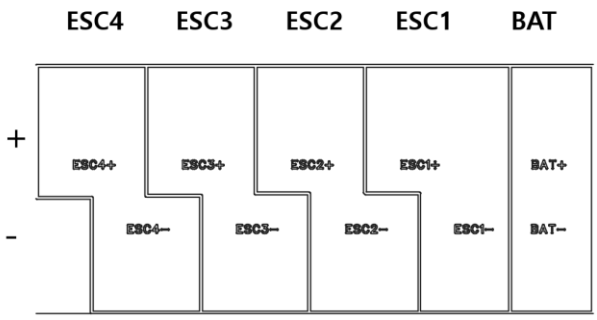
UNIT: mm

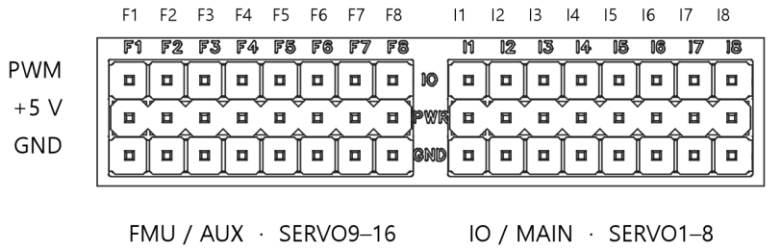
Dimensions in mm refer to the enclosure, excluding protruding connectors and pins. The 72 × 114 mm hole pattern is for enclosure assembly.

Secure the FC to the aircraft with double-sided tape. Allow clearance for USB, microSD, connectors and protruding pins.

Power connections and PWM outputs

Top view with BAT/ESC at the top. BAT is the battery input; ESC1–4 are power pads for external ESCs.





IO PWM OUT = MAIN OUT	I1~I8 → SERVO1~8
FMU PWM OUT = AUX OUT	F1~F8 → SERVO9~16

The PWM PWR row supplies 5 V shared by IO and FMU outputs. Do not connect another BEC in parallel. Input supports 3–6S LiPo, up to 25.2 V (6S).

Battery monitor settings

Mission Planner → CONFIG → Full Parameter List

Voltage multiplier	BATT_VOLT_MULT	8.987575
Current scale (optional)	BATT_AMP_PERVLT	0.25 A/V
Current offset	BATT_AMP_OFFSET	0 V

BATT_MONITOR = 3 (default: voltage only) | BATT_VOLT_PIN = 8

To view FC supply current, optionally set BATT_MONITOR=4 and BATT_CURR_PIN=4, then reboot. Use the listed scale and offset and calibrate against a meter.

The sensor measures only the FC supply branch, excluding ESC propulsion current. Do not use it for total aircraft consumption, remaining capacity or capacity-based failsafes. Use a separate current sensor for total battery monitoring.

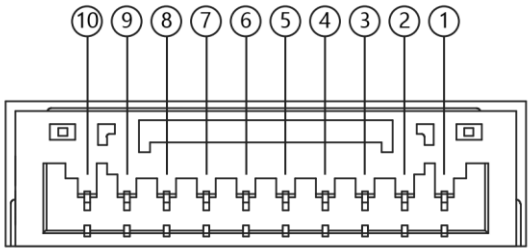
Saved settings survive updates. To use voltage-only monitoring, save BATT_MONITOR=3 and reboot.

GPS ports

Top view with Front at the top. Circled numbers identify pins. Left and right are reversed when viewed from the cable plug mating face.

GPS1

SERIAL3 / USART1

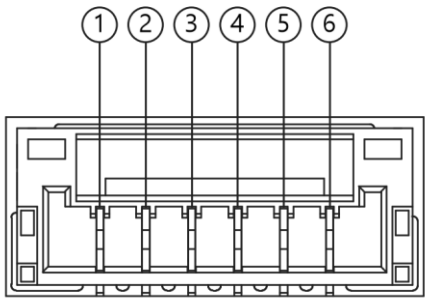


10 pins / 1.25 mm pitch

Pin	Signal
1	GND
2	BUZZER
3	+3.3 V
4	SAFETY LED
5	SAFETY SW
6	SDA
7	SCL
8	RX
9	TX
10	+5 V

GPS2

SERIAL4 / UART8



6 pins / 1.25 mm pitch

Pin	Signal
1	+5 V
2	TX
3	RX
4	SCL
5	SDA
6	GND

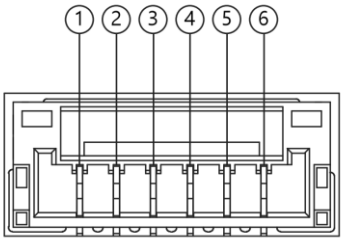
TX/RX are named from the FC perspective. Connect FC TX → device RX, FC RX ← device TX, GND ↔ GND, SCL ↔ SCL and SDA ↔ SDA. Check the peripheral power requirements.

Telemetry and I2C ports

Top view with Front at the top. Circled numbers identify pins. Left and right are reversed when viewed from the cable plug mating face.

TLM1

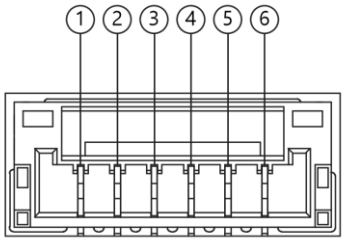
SERIAL1 / UART7



Pin	Signal
1	+5 V
2	TX
3	RX
4	CTS
5	RTS
6	GND

TLM2

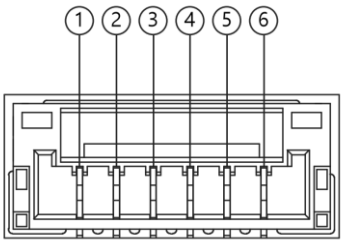
SERIAL2 / UART5



Pin	Signal
1	+5 V
2	TX
3	RX
4	CTS
5	RTS
6	GND

TLM3

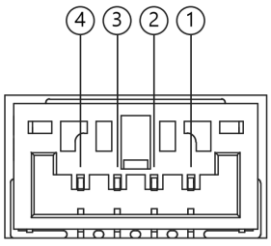
SERIAL5 / USART2



Pin	Signal
1	+5 V
2	TX
3	RX
4	SCL
5	SDA
6	GND

I2C

I2C4 / 1.25 mm pitch



Pin	Signal
1	GND
2	SDA
3	SCL
4	+5 V

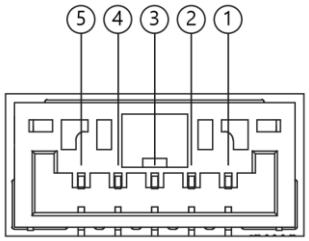
TLM1 and TLM2 have CTS/RTS. TLM3 pins 4/5 are SCL/SDA. The 5 V pin supplies peripheral power. Do not apply 5 V power to UART or I2C signal pins.

Receiver and service ports

Top view with Front at the top. Circled numbers identify pins. Left and right are reversed when viewed from the cable plug mating face.

S.B

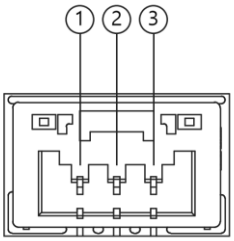
SBUS RC INPUT



Pin	Signal
1	GND
2	NC
3	RSSI
4	SBUS IN
5	+5 V

S.OUT

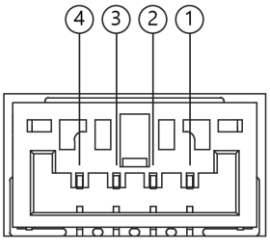
IO USART3 TX / OUTPUT



Pin	Signal
1	NC
2	SIGNAL OUT
3	GND

FUUP

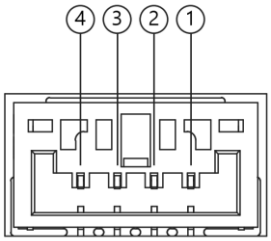
FMU SWD / SERVICE



Pin	Signal
1	3.3 V REF
2	SWCLK
3	SWDIO
4	GND

IOUP

IO SWD / SERVICE



Pin	Signal
1	3.3 V REF
2	SWCLK
3	SWDIO
4	GND

S.B is the SBUS input; S.OUT carries the IO USART3 TX signal. S.OUT has no power output pin.

Leave NC pins unconnected. FUUP and IOUP are SWD service ports; 3.3 V REF is the debugger voltage reference.